

Moisture recycling: when land use changes rain somewhere else

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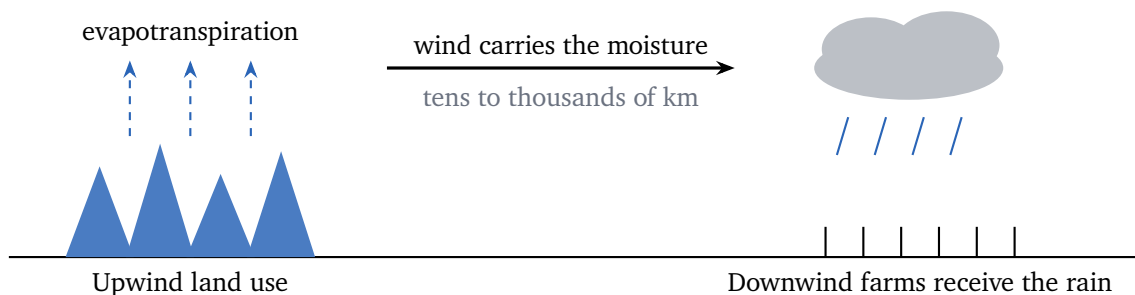
This note is an entry point into the literature on moisture recycling and land use, and where it stands today. It grows out of my research on how land-use change alters rainfall and temperature, in India¹ and West Africa,² and my current World Bank work with Esha Zaveri on valuing forest conservation.

Disclaimer: not an expert's verdict, but a clarification for anyone who wants to take up this literature. Comments and corrections from experts are very welcome.

The idea

Rain over land does not come only from the ocean. Plants and soils return water to the air through evapotranspiration, winds carry that moisture, and part of it falls again as rain further away. Once the flows are reconciled with observed water budgets, about **45% of the rain falling on land** evaporated from land first. Forests are the largest single source.

The economic consequence is simple to state. If a forest upwind is cleared, or a desert upwind is irrigated, the rain received by a farmer downwind changes. That farmer did not take the decision and is not compensated for it. Moisture recycling is an **atmospheric externality** of land use, and it crosses watersheds and national borders.



What clearing a forest does to rain

A few effects are worth keeping in mind before reading any single number. They depend on **distance**, **season** and **scale**. Near a clearing, rain can rise in the wet season, because the bare surface heats up and triggers convection. Beyond about 50 km, and everywhere in the dry season, rain falls (Smith et al., 2023; Qin et al., 2025). The response is also not linear: moderate clearing can raise rainfall before heavy clearing lowers it (Khanna et al., 2017; Leite-Filho et al., 2021).

Not only forests

Any land use that changes evaporation changes rain downwind. **Irrigation** works in the opposite direction: it adds moisture and cools, and its downwind benefits on the US Great Plains are as large as the on-farm ones. **Cropland**, **savanna**, **Sahel vegetation** and **urban areas** matter too. This links *blue water* (rivers, aquifers), which is metered and governed, to *green water*, the moisture returned to the air by soils and plants, which represents two-thirds of rainfall and needs to be accounted for in the economic valuation of ecosystems.

¹Degois, M. (2026). Irrigation Makes It Rain? Climate Spillovers and Adaptation in India. Job market paper.

²Degois, M., Vendryes, T. and Wolfersberger, J. (2025). Deforestation, Downwind Rainfall and Conflict. Working paper.

How we measure it (to date)

Two families of methods answer different questions. **Physical models** ask where evaporated water goes, which is what you need to trace moisture across a border. **Empirical studies** ask what happened to rainfall when land cover actually changed, which is what gives you a confidence interval.

Approach	Examples	What it does	Main limit
Lagrangian tracking	UTrack, FLEXPART, WaterSip, QIBT	Follows air parcels from where water evaporates to where it rains out, through reanalysis winds	Sensitive to vertical mixing; the public UTrack dataset is a fixed 2008–17 average
Eulerian tracking	WAM2layers, DRM	Solves a tagged water budget on a grid	Few vertical layers; tends to attribute rain to more distant sources
Online tagging in a climate model	WRF water-vapour tracers	Tags moisture inside a running model, so the atmosphere can respond to the land change	Expensive, regional, model-dependent
Ensembles and reconciled flows	AMSSRAB; De Petrillo et al.	Combine several trackers, or rebalance tracked flows against observed water budgets	Coarse (1°) or country level only; agreement is poor in drylands
Empirical, wind-based designs	Satellite rainfall and forest data; economics papers	Build each place's exposure to upwind land-use change from winds or trajectories, then compare places with fixed effects	Needs credible variation in exposure; units differ across papers

Tracking models.

- + They trace moisture from source to sink, across borders, with global coverage, and give flows in millimetres, so a recycling share can be computed for any place.
- Almost all of them are offline: they take the atmosphere as given and cannot represent how the circulation responds to the land change. This response drives, for instance, the rainfall increase right next to a clearing, or the weaker monsoon over irrigated land.
- None computes evaporation. Each takes an evapotranspiration input (ERA5, GLEAM, FLUXCOM), and these inputs, like the trackers themselves, disagree most in drylands.

Empirical estimates.

- + They measure what actually happened, including the circulation response, come with confidence intervals, and link directly to economic outcomes. Built on wind or moisture trajectories, they also trace where the effect comes from.
- They need credible variation in exposure, rely on noisy satellite rainfall, and are local or regional, so they are hard to extrapolate.
- Their units (forested pixels upwind, wind-exposure indices) differ across papers, so they are hard to compare with each other or with tracking models.

Economic valuation of the externality

The economics literature is small: about a dozen studies put a value on the effect. Most follow the same three steps: measure exposure to upwind land-use change, estimate its effect on rainfall or temperature, then trace it to an economic outcome.

- **Hydropower.** Upwind deforestation lowers generation at Brazilian dams by 2.5 to 10% (Araujo, 2024).
- **Agriculture.** A US tree-planting programme of the 1930s raised rainfall up to 200 km downwind for

decades and helped small farms survive (Grosset-Touba, Papp and Taylor). Irrigation cooling on the Great Plains is worth about US\$120 million a year in downwind crop yields (Braun and Schlenker, 2023).

- **Conflict.** In West Africa, upwind deforestation lowers rainfall up to 200 km downwind and raises conflict, mainly civil unrest, for up to six years (Degois, Vendryes and Wolfersberger).
- **Adaptation, medium and long run.** In India, irrigation raises dry-season rainfall downwind. Farmers adapt by expanding private irrigation and double cropping, and in the long run larger farms expand, polarizing the agricultural sector (Degois, job market paper).
- **Value of the service.** Each hectare of Amazon forest generates rainfall worth about US\$59 a year at Brazil's agricultural water tariff, about US\$20 billion a year for the Legal Amazon (Baker et al., 2026). A recent report estimates that 30% of Paraguay's economy depends on rain generated in the Amazon (Ranger et al., 2026).

Standard forest valuation still leaves this service out: a meta-analysis of 261 studies and 758 valuations of forest ecosystem services does not include recycled rainfall at all (Taye et al., 2021, *Ecological Economics*).

Identified gaps, with plenty of research ahead

These are the objectives of my current report for the World Bank with Esha Zaveri.

1. **Reconcile tracking models and empirical estimates**, and understand where they disagree. Six leading global models get the sign of the soil moisture to rainfall feedback wrong at 50 to 100 km (Taylor et al., 2012), and satellites consistently show larger deforestation effects than models.
2. **Value more economic outcomes, and the cost of conservation.** Existing values use one water price and a linear response, and most coefficients rest on wind-based exposure indices rather than quantified moisture flows. Policy makers also need the cost of keeping the forest standing, not only the value of what it provides.
3. **Link land-use changes to each other through adaptation.** Forest loss lowers rain, which raises irrigation, which changes rain elsewhere. The forest and irrigation literatures measure two legs of this circuit but have not been joined.
4. **Identify winners and losers.** No study yet asks who bears the loss and who gains.

But also:

- **Geography.** The evidence is thinnest where dependence is highest: Latin America holds nearly half of the regional studies, Sub-Saharan Africa about a dozen, the Middle East and North Africa none. South Asia, East Asia and Europe have no economic study.
- **Drylands.** Trackers and evaporation inputs disagree most in arid zones, where dependence on recycling is highest.
- **Other land-use change.** Cropland, savanna, non-forest vegetation and urban areas are far less studied than forests.
- **Seasonality.** Effects reverse between seasons, yet most estimates are annual.
- **Nonlinearity.** Most coefficients are linear; several studies find thresholds, asymmetries and sign reversals.
- **Short run versus long run, and adaptation.** Most estimates are short run, and few follow how people adapt. None is general equilibrium.

Where my work fits. My job market paper studies the irrigation leg. It is the first empirical evidence of how irrigation modifies the climate differently across seasons, and it traces the consequences for adaptation and long-run agricultural development in India. With Thomas Vendryes and Julien Wolfersberger, I study the forest leg in West Africa, where upwind deforestation lowers rainfall and raises conflict. At the World Bank, with Esha Zaveri, I work on the first three objectives above, to give policy makers numbers they can use and to bridge economic estimates and the calibration of climate models.

Where to start reading

Reviews

- Eltahir and Bras (1996). Precipitation recycling. *Reviews of Geophysics* 34.
- te Wierik et al. (2021). Reviewing the impact of land use and land-use change on moisture recycling and precipitation patterns. *Water Resources Research* 57.

Datasets

- Tuinenburg, Theeuwes and Staal (2020). High-resolution global atmospheric moisture connections from evaporation to precipitation (UTrack). *Earth System Science Data* 12.
- Zhang et al. (2025). A multi-model based dataset of global atmospheric moisture source-sink relationships and atmospheric basins (AMSSRAB). *Scientific Data*.

Forest cover and forest clearing

- Shukla, Nobre and Sellers (1990). Amazon deforestation and climate change. *Science* 247.
- Smith, Baker and Spracklen (2023). Tropical deforestation causes large reductions in observed precipitation. *Nature* 615.
- Qin et al. (2025). Impact of Amazonian deforestation on precipitation reverses between seasons. *Nature* 639.

Irrigation

- Alter, Im and Eltahir (2015). Rainfall consistently enhanced around the Gezira Scheme in East Africa due to irrigation. *Nature Geoscience* 8.
- Greve et al. (2025). Observational evidence of increased afternoon rainfall downwind of irrigated areas. *Nature Communications* 16.
- Agrawal et al. (2019). Effects of winter- and summer-time irrigation over the Gangetic Plain on the mean and intra-seasonal variability of the Indian summer monsoon. *Climate Dynamics* 53.

Other land-use change

- Loarie et al. (2011). Direct impacts on local climate of sugar-cane expansion in Brazil. *Nature Climate Change* 1.
- Li et al. (2023). Increased precipitation over land due to climate feedback of large-scale bioenergy cultivation. *Nature Communications* 14.

In economics

- Araujo (2024). The value of tropical forests to hydropower. *Energy Economics* 129.
- Araujo. When clouds go dry. Working paper.
- Grosset-Touba, Papp and Taylor. Rain follows the forest. Working paper.
- Braun and Schlenker (2023). Cooling externality of large-scale irrigation. NBER Working Paper 30966.